

Multiple Choice (10 marks)

1. Let A be the set of all ordered pairs of integers (m, n) such that $7m + 12n = 22$. What is the greatest negative number in the set $B = \{m + n : (m, n) \in A\}$?
 - (a) -5
 - (b) -4
 - (c) -3
 - (d) -2
2. For how many positive integers k does the ordinary decimal representation of the integer $k!$ end in exactly 99 zeros?
 - (a) None
 - (b) One
 - (c) Four
 - (d) Five
3. Find the maximum possible order for some element of $\mathbb{Z}_8 \times \mathbb{Z}_{10} \times \mathbb{Z}_{24}$.
 - (a) 8
 - (b) 120
 - (c) 240
 - (d) 24
4. Find the generator for the finite field $\mathbb{Z}_{11}$.
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
5. Let $(\mathbb{Z}_{10}, +, \cdot)$ be the ring of integers modulo 10, and let S be the subset of $\mathbb{Z}_{10}$ represented by $\{0, 2, 4, 6, 8\}$. Which of the following statements is FALSE?
 - (a) $(S, +, \cdot)$ is closed under addition modulo 10.
 - (b) $(S, +, \cdot)$ is closed under multiplication modulo 10.
 - (c) $(S, +, \cdot)$ has an identity under addition modulo 10.
 - (d) $(S, +, \cdot)$ has no identity under multiplication modulo 10.

True / False (10 marks)

Answer True or False

6. True or False: There are exactly 3 nonisomorphic trees that can be formed with 5 vertices.
7. True or False: The only coset of the subgroup $4\mathbb{Z}$ in $2\mathbb{Z}$ is $4\mathbb{Z}$.
8. True or False: The product $(20)(-8)$ in $\mathbb{Z}_{26}$ is equal to 22.
9. True or False: The largest order of an element in the group of permutations of 5 objects is 6.
10. True or False: The probability that 0 will not be chosen when selecting k digits from $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ is $(9/10)^k$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Given a real 2×2 matrix A such that $A = A^{(-1)}$ and A is distinct from the identity matrix and its negative, analyze the implications on the trace of A . Justify your reasoning.
12. Analyze the expression $x + 4z$ as a function of variables in $\mathbb{R}^3$, and determine its minimum value given the constraint $x^2 + y^2 + z^2 \leq 2$. Provide a detailed explanation of your solution.

End of Paper